Dictionaries

→ Unordered → key-value pairs
→ keys should be ~~must~~ immutable

(A) using the same key again overwrites the previous key's value.

(B) → get() method.

eg → a = {'student': 'Krish', 'grade': 'B'}

a.get('student') → returns 'Krish'

→ we can pass a default value in get(), if passed key is not found.

(A) eg. a.get('age', 'Unavailable')

→ ~~prints~~ returns unavailable.

(A) del a['grade']

deletes the key value pair of 'grade'.

(A) dict.keys() → returns all keys in the dict

(A) dict.values() → returns all the values in the dict.

(A) dict.items() → returns all the key-value pairs in the form of tuples.

$a = \{'name' = 'Krishi'\}$

Shallow copy (~~copy~~ dict.copy()

when we make a copy,
e.g. ~~a=~~ $b = a$.
and update $a['name'] = 'Max'$

then $b['name']$ also $= 'Max'$.

* Note: mutable objects variables when copied * reference to the same object in memory.

Hence to make a copy,

$\boxed{b = a.copy()}$ # shallow copy

Dictionary comprehension

$\boxed{squares = \{x : x**2 \text{ for } x \text{ in range}(5)\}}$

$\Rightarrow \{0:0, 1:1, 2:4, 3:9, 4:16\}$

We can use conditions,

$\boxed{even_sq = \{x : x**2 \text{ for } x \text{ in range}(4) \text{ if } x\%2==0\}}$

$\Rightarrow \{0:0, 2:4, 4:16\}$

Merge dictionaries

$a = \{'a':1\}$; $b = \{'b':2\}$

$\boxed{m = \{**a, **b\}}$

$\Rightarrow m = \{'a':1, 'b':2\}$

Note: $**x$
key ~~value~~ word argument
(for any value present in form of key value pairs)